

CS173: Discrete Structures

Problem Set 0

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1. Determine the truth value, if any, of each of the following sentences.

- (a) “Every integer is either even or odd.”

TRUE.

Let n be an integer. Thus $2n$ is an even number, and $2n + 1$ is and an odd number. Incrementing up from $2n + 1$ is $2n + 2 = 2k$ where $k = n + 1$ and is an integer meaning $2k$ is again even. By the principles of induction all integer are thus even or odd.

- (b) “Every number is either even or odd.”

FALSE.

Can be proven by counterexample. 2.5 is a number and is neither even nor odd.

- (c) “Every non-empty set of integers has a smallest element.”

FALSE.

Can be proven by counterexample. Take the set of all negative integers $\{\dots, -2, -1\}$. This set has no smallest element.

- (d) “Every non-empty set of natural numbers has a smallest element.”

TRUE.

The set of natural numbers is bounded below by 0 hence all subsets of natural numbers must have a smallest value.

- (e) “Every natural number is boring.”

FALSE.

The premise of being boring is dialectical. Fore there to be 'boring' natural numbers there has to an opposite, or a reference. That suggest that there must exist some 'interesting' or 'non-boring' natural numbers for 'boring' natural numbers to exist. If all numbers were boring, then none would be boring. The term would lose meaning.

- (f) “No natural number is boring.”

FALSE.

From part (e), there has to exist some boring natural numbers so that non-boring' natural numbers exist.

2. Determine the truth value, if any, of each of the following sentences.

- (a) “This sentence is true.”

INDETERMINATE.

Let $S :=$ “ S is true.” If S is true, then the assertion is true, which is consistent. If S is false, then what S assertion is false, which is also consistent.

- (b) “This sentence is false.”

PARADOX.

Let $Q :=$ “ Q is false.” If Q is true, then what Q asserts must hold, meaning Q is false which is a contradiction. If Q is false, then what Q asserts does not hold, meaning Q is not false, so Q is true which again is a contradiction.

- (c) “The set of all sets that contain themselves contains itself.”

INDETERMINATE.

Let this set be R . If $R \in R$, then R satisfies the definition of R and thus is in R , which is consistent. If $R \notin R$, then R does not satisfy the definition, which is also consistent.

- (d) “The set of all sets that do not contain themselves does not contain itself.”

PARADOX.

Let this set be N . If $N \notin N$, then N satisfies the definition of N and thus is in N , which is a contradiction. If $N \in N$, then N does not satisfy the definition, but is in N which is also a contradiction.

- (e) “If this sentence is false, then 7 is a prime number.”

TRUE.

Let $A :=$ “If A is false, then 7 is a prime number.” and $B :=$ “7 is prime”. Then $A := \neg A \Rightarrow B$. If A is false then $\neg A$ is true, which implies B has to be true. B is obviously true, which means this is consistent. If A is true then $\neg A$ is false, which means B is not dependent on A and can be either $\top$ or $\perp$, thus this is still consistent.

- (f) “If this sentence is true, then 7 is not a prime number.”

FALSE.

Let $A :=$ “If A is true, then 7 is not a prime number.” and $B :=$ “7 is not prime”. Then $A := A \Rightarrow B$. If A is true then it implies B has to be true. But B is obviously false, so this is inconsistent. If A is false then it means B is not dependent on A and can be either $\top$ or $\perp$, thus this is consistent. So A has to be $\perp$.

3. Consider the infinite sequence of sentences indexed by natural numbers

$$S_0, S_1, S_2, \dots, S_i, \dots$$

where each sentence asserts that all of the subsequent sentences are false.

$$S_i := \text{“}S_j \text{ is false for all } j > i\text{”}$$

For example, $S_3 := \text{“}S_j \text{ is false for all } j > 3\text{”}$ which means S_3 is equivalent to the assertion that “the sentences $S_4, S_5, S_6, \dots$ are all false.”

What are the truth values, if any, of the sentences in this sequence?

All S_n are INDETERMINATE.

S_0 cannot be $\top$, because if S_0 were true, then it would assert that S_1 is $\perp$. But S_1 says that all sentences S_n with $n > 1$ are false. If S_1 is false, then there exists some S_n with $n > 1$ that is true, which contradicts the assertion made by S_0 that all subsequent sentences are false.

The same applies to any S_n : assuming S_n is $\top$ forces S_{n+1} to be $\perp$, which in turn implies the existence of some later sentence that is $\top$, contradicting the original claim of S_n .

The only way all sentences could avoid contradiction by being true would be if there were a final sentence S_N that is true. However, there is no last sentence, since the sequence is infinite.

Therefore, none of the sentences S_n can be assigned a truth value.